

1.5 Chemical Energetics

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.5 Chemical Energetics
Difficulty	Easy

Time allowed: 30

Score: /20

Percentage: /100

Question 1a

Standard enthalpy changes include enthalpy of formation, enthalpy of combustion and enthalpy of neutralisation.

i)

Define *enthalpy of formation*.

[1]

ii)

Write an equation that represents the enthalpy of formation of ammonia, NH_3 .

[1]

[2 marks]

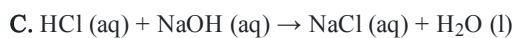
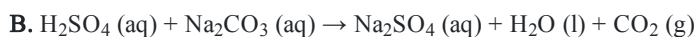
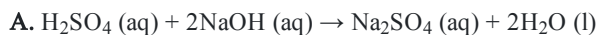
Question 1b

State what is meant by *standard* conditions in enthalpy measurements.

[1 mark]

Question 1c

Which of the following equation(s) can represent the enthalpy of neutralisation?



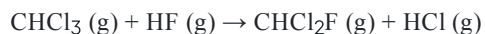
Explain your answer.

[2 marks]

Question 2a

Freon 21 is a hydrochlorofluorocarbon that was used as a propellant and refrigerant but later withdrawn from use because of its effect on ozone depletion. The formula is CHCl_2F .

Freon 21 is produced by reacting hydrogen fluoride with trichloromethane in the following reaction:



Use the data in Table 2.1 to calculate the enthalpy of reaction, ΔH_r , for the formation of $\text{CHCl}_2\text{F}(\text{g})$.

compound	enthalpy change of formation, $\Delta H_f^\circ / \text{kJ mol}^{-1}$
$\text{CHCl}_3(\text{g})$	-103.2
$\text{CHCl}_2\text{F}(\text{g})$	-284.1
$\text{HF}(\text{g})$	-273.3
$\text{HCl}(\text{g})$	-92.3

Table 2.1

enthalpy change of reaction, $\Delta H_r = \dots\dots\dots \text{kJ mol}^{-1}$

[2 marks]

Question 2b

Using Fig. 2.1, draw a labelled reaction pathway diagram for part (a) and include the activation energy.

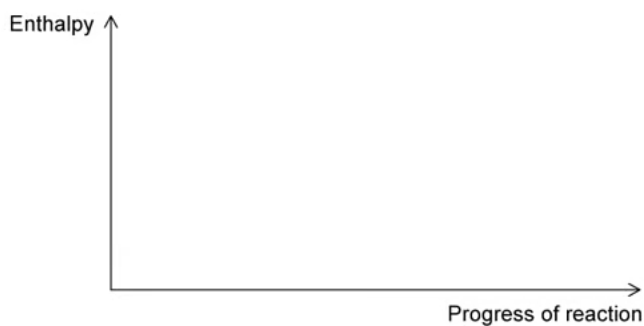


Fig. 2.1

[2 marks]

Question 2c

The reaction pathway diagram has a feature known as the **transition state**. Explain what this means.

[1 mark]

Question 3a

Define the term average bond enthalpy.

[2 marks]

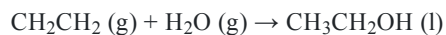
Question 3b

Give the formula for calculating the standard enthalpy change of reaction, ΔH_r , using bond energies.

[1 mark]

Question 3c

Use the data given in Table 3.1 to calculate the enthalpy change for the hydration of ethene to form ethanol.

**Table 3.1**

Bond	Bond energy / kJ mol^{-1}
C-C	346
C=C	614
C-H	414
C-O	358
O-H	463

[3 marks]**Question 3d**

Use Table 3.2 to calculate the following enthalpy change for the reaction of ethane and bromine to form 1,2-dibromoethane and hydrogen bromide.

Table 3.2

Bond	Bond enthalpy / kJ mol^{-1}
C-C	346
C-H	414
C-Br	285
H-Br	366
Br-Br	193

[4 marks]

Model Answers: Easy

1a

a)

i) The definition of *enthalpy of formation* is:

- The enthalpy change when 1 mole of a compound is formed from its elements under standard conditions; [1 mark]

ii) The equation for the enthalpy formation of ammonia is:

- $\frac{1}{2}\text{N}_2(\text{g}) + 1\frac{1}{2}\text{H}_2(\text{g}) \rightarrow \text{NH}_3(\text{g})$; [1 mark]

[Total: 2 marks]

- The three underlined terms in part (i) are essential to scoring the mark
- Careful:** This equation often trips people up
 - You will be accustomed to seeing this equation with integers as the coefficients (1,3,2)
 - This would be incorrect for an equation that describes enthalpy of formation as it does not produce 1 mole of the product

1b

b) What is meant by *standard* conditions in enthalpy measurements:

- The temperature must be 298 K
AND
The pressure must be 101 kPa; [1 mark]

[Total: 1 mark]

- You need to know the standard conditions
- By definition, the word standard is describing the conditions so the answer requires 298 K and 101 kPa

1c

c) The equation(s) which can represent the enthalpy of neutralisation are:

- **B and C**; [1 mark]
- (Because) the enthalpy of neutralisation produces 1 mole of water from an acid and base; [1 mark]

[Total: 2 marks]

- The identity of the acid and base are irrelevant as long as only one mole of water is produced in the reaction
- Don't be put off by reactions with carbonates and CO_2 as a product - it is still a neutralisation reaction
- An acid-base reaction does not necessarily have to be neutralisation
 - For example, when ammonia reacts with hydrogen chloride, the reaction is acid-base, but not neutralisation:
 - $\text{HCl} + \text{NH}_3 \rightleftharpoons \text{NH}_4\text{Cl}$

2a

a) To calculate the enthalpy of reaction, ΔH_r , for the formation of $\text{CHCl}_2\text{F(g)}$:

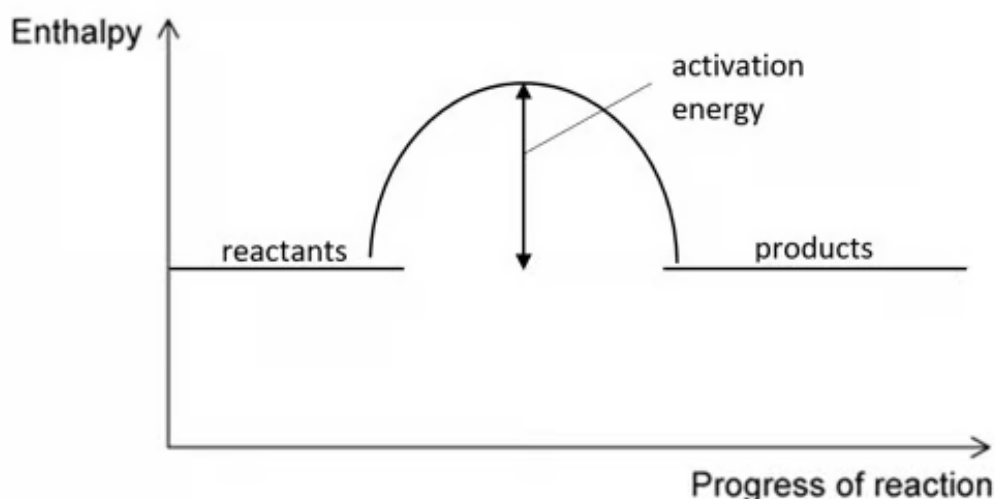
- $\Delta H_r = (-284.1) + (-92.3) - (-103.2) - (-273.3)$; [1 mark]
- $0.1 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]

[Total: 2 marks]

- To solve this problem you need to use the formula:
 - $\Delta H_r = \sum \Delta H_{f \text{ products}} - \sum \Delta H_{f \text{ reactants}}$
- Be careful not to mix up the signs of the energy terms with the mathematical operations
 - **Tip:** Use brackets around the terms
- If there are coefficients >1 in the balanced equation, don't forget to multiply the energy terms by the coefficient

2b

b) A reaction pathway diagram for part (a) with activation energy labelled is:



- Correct diagram with the correct positions of reactants and products ; [1 mark]
- Activation energy correctly labelled; [1 mark]

[Total: 2 marks]

- You can use the formulae of the reactants and products instead of the labels 'reactants' and 'products'
- The enthalpy of reaction is close to zero, so the two horizontal lines should be at an almost identical height
- If the lines are far apart then you will not score the mark as you have not applied correctly the information calculated in part (a).

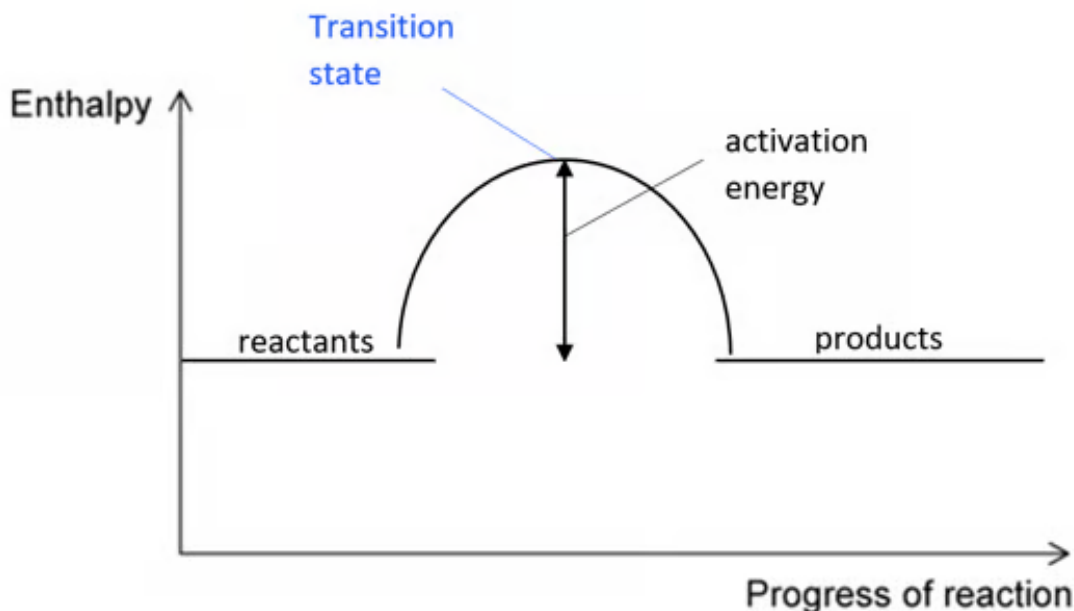
2c

c) Explain what is meant by the transition state:

- The transition state is a stage during the reaction at which chemical bonds are partially broken and formed; [1 mark]

[Total: 1 mark]

- The transition state is very unstable – it cannot be isolated and is higher in energy than the reactants and products
- It is located at the very top of the curve on the reaction pathway diagram.



3a

a) The definition of average bond enthalpy is:

- The energy needed to break one mole of bonds; [1 mark]
- In a gaseous molecule averaged over similar compounds; [1 mark]

[Total: 2 marks]

- The average bond enthalpy term is the average amount of energy needed to break a specific type of bond, measured over a wide variety of different gaseous molecules
- It is essentially the average of all the bond dissociation enthalpies for a specific type of bond

3b

b) The formula for calculating the standard enthalpy change of reaction, ΔH_r , using bond energies is:

- $\Delta H_r = \text{enthalpy change for bonds broken} + \text{enthalpy change for bonds formed}$

OR

$$\Delta H_r = \sum \Delta H \text{ Bonds broken} + \sum \Delta H \text{ Bonds made; [1 mark]}$$

[Total: 1 mark]

- Bond energies are used to find the $\Delta H_r^\ominus$ of a reaction when this cannot be done experimentally

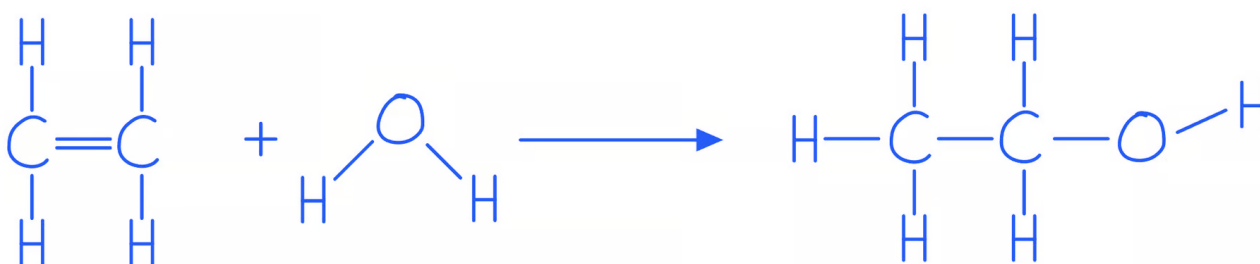
3c

c) To calculate the enthalpy change for the hydration of ethene:

- Σ bonds broken (endothermic)
 $E(\text{C}=\text{C}) = 614$
 $E(\text{C}-\text{H}) \times 4 = 1656$
 $E(\text{O}-\text{H}) \times 2 = 926$
Total = 3196 kJ; [1 mark]
- Σ bonds made (exothermic)
 $(\text{C}-\text{C}) = 346$
 $E(\text{C}-\text{H}) \times 5 = 2070$
 $(\text{C}-\text{O}) = 358$
 $(\text{O}-\text{H}) = 463$
Total = -3237 kJ; [1 mark]
- $\Delta H_r = 3196 + (-3237) = -41 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 3 marks]

- It can be really helpful to draw out the structures of the reactants and products when calculating enthalpy change using bond enthalpies as it is easier to see the number of each type of bond

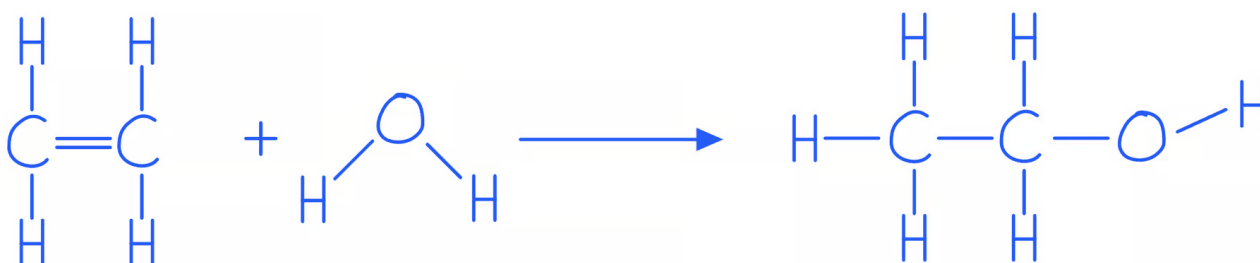


3d

- Σ bonds made (exothermic)
 $(\text{C-C}) = 346$
 $E(\text{C-H}) \times 5 = 2070$
 $(\text{C-O}) = 358$
 $(\text{O-H}) = 463$
 Total = -3237 kJ; [1 mark]
- $\Delta H_r = 3196 + (-3237) = -41 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 3 marks]

- It can be really helpful to draw out the structures of the reactants and products when calculating enthalpy change using bond enthalpies as it is easier to see the number of each type of bond



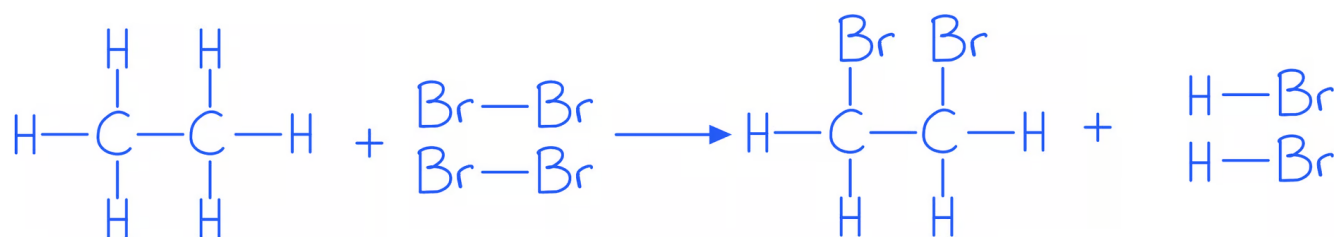
3d

d) To calculate the enthalpy change for the reaction of ethane and bromine:

- $\text{CH}_3\text{CH}_3 + 2\text{Br}_2 \rightarrow \text{CH}_2\text{BrCH}_2\text{Br} + 2\text{HBr}$; [1 mark]
- Σ bonds broken (endothermic)
 $E(\text{C-C}) = 346$
 $E(\text{C-H}) \times 6 = 2484$
 $E(\text{Br-Br}) \times 2 = 386$
 Total = 3216 kJ; [1 mark]
- Σ bonds formed (exothermic)
 $E(\text{C-C}) = 346$
 $E(\text{C-H}) \times 4 = 1656$
 $E(\text{C-Br}) \times 2 = 570$
 $E(\text{H-Br}) \times 2 = 732$
- Total = -3304 kJ; [1 mark]
- $\Delta H_r = -88 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 4 marks]

- Again, drawing out the correct structures will help here given that there are two molecules of bromine and hydrogen bromine



Cl-Cl	242
H-Cl	431

Include a sign in your answer.

enthalpy change =
..... kJ mol⁻¹

[3]

- ii) State the conditions needed for this reaction to occur.

[1]

How did you do?



Stuck? [View related notes](#)

[View Answer](#)

Questions

Structured Questions

0%

1.6... 0%



1.7... 0%



1.8 ... 1%



2. Inor... 0%



3. Orga... 1%



4. Anal... 0%

